DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-200-04 — Ultrasonic thickness survey of 6"-P300-P-1011

Area	A200 Vessels and column	**Task**	UT Ultrasonic thickness survey
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	**C-101** — C-101 vertical vessel 48" ID × 240" T/T — E 68'-6" N 25'-0" EL 100'-8-1/4"		
Equipment	C-101	**Lines**	6"-P300-P-1011
Isometrics	ISO-6"-P300-P-1011 (generated); owner's drawings ISO-P-1011		
Permits	General work permit		

Reason

Thickness survey of the column overhead line.

Scope of work

1. Erect access and remove insulation at the TMLs where fitted (class H300).
2. Calibrate the thickness gauge on a step block covering 0.21875 in.
3. Take four readings 90° apart at each of the 15 TMLs (SW-WO-6).
4. Report readings against the owner's minimum wall; raise a spool replacement for any reading below it.

Affected items: 6"-P300-P-1011/el8, 6"-P300-P-1011/p9, 6"-P300-P-1011/el10, 6"-P300-P-1011/p11, 6"-P300-P-1011/el12, 6"-P300-P-1011/p13, 6"-P300-P-1011/T:3/4"-P300-I-7011, 6"-P300-P-1011/el16, 6"-P300-P-1011/el18, 6"-P300-P-1011/p19, 6"-P300-P-1011/el20, 6"-P300-P-1011/p21, 6"-P300-P-1011/el22, 6"-P300-P-1011/p23, 6"-P300-P-1011/el24

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-WO-1, SW-WO-2, SW-WO-6, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1011	6"-P300-P-1011	6"-P300-P-1011	2	0 error(s), 3 warning(s) — see below

ISO-P-1011 (PDF) — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1011	2	warning	W14 (6"-P300-P-1011/W14, sheet 2) is labelled again on sheet 1; it is 162.0 in from the nearest break
ISO-P-1011	2	warning	weld 6"-P300-P-1011/W20 (E 89'-0" N 25'-0" EL 110'-2 1/2", sheet 2) has no W20 label
ISO-P-1011	1	warning	branch weld 3/4"-P300-I-7011/W1 (E 68'-11" N 6" EL 120'-0", sheet 1) has no W21 label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
C-101	equipment	C-101 vertical vessel 48" ID × 240" T/T		E 68'-6" N 25'-0" EL 100'-8-1/4"
6"-P300-P-1011/el8	item	ELBOW 90° R = 5/3 OD 6" P300	6"-P300-P-1011	E 68'-6" N 24'-7" EL 129'-7"
6"-P300-P-1011/p9	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-P-1011	E 68'-6" N 18'-6" EL 130'-0"
6"-P300-P-1011/el10	item	ELBOW 90° R = 5/3 OD 6" P300	6"-P300-P-1011	E 68'-6" N 12'-5" EL 129'-7"
6"-P300-P-1011/p11	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-P-1011	E 68'-6" N 12'-0" EL 125'-0"
6"-P300-P-1011/el12	item	ELBOW 90° R = 5/3 OD 6" P300	6"-P300-P-1011	E 68'-6" N 11'-7" EL 120'-5"
6"-P300-P-1011/p13	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-P-1011	E 68'-6" N 6'-1/2" EL 120'-0"
6"-P300-P-1011/T:3/4"-P300-I-7011	item	TEE REDUCING 6" × 3/4" P300	6"-P300-P-1011	E 68'-6" N 6" EL 120'-0"
6"-P300-P-1011/el16	item	ELBOW 90° R = 5/3 OD 6" P300	6"-P300-P-1011	E 68'-6" N -7" EL 119'-7"
6"-P300-P-1011/el18	item	ELBOW 90° R = 5/3 OD 6" P300	6"-P300-P-1011	E 68'-11" N -12" EL 117'-5"
6"-P300-P-1011/p19	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-P-1011	E 78'-9" N -12" EL 117'-0"
6"-P300-P-1011/el20	item	ELBOW 90° R = 5/3 OD 6" P300	6"-P300-P-1011	E 88'-7" N -12" EL 116'-7"
6"-P300-P-1011/p21	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-P-1011	E 89'-0" N -12" EL 114'-9"
6"-P300-P-1011/el22	item	ELBOW 90° R = 5/3 OD 6" P300	6"-P300-P-1011	E 89'-0" N -7" EL 112'-11"

Tag	Kind	Description	Line	Location
6"-P300-P-1011/p23	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-P-1011	E 89'-0" N 12'-0" EL 112'-6"
6"-P300-P-1011/el24	item	ELBOW 90° R = 5/3 OD 6" P300	6"-P300-P-1011	E 89'-0" N 24'-7" EL 112'-1"

Inspections

Id	Inspection	Criteria	Basis
WO-200-04-I1	Ultrasonic thickness at 15 TMLs, four readings each	Nominal wall 0.21875 in; each reading above the owner's minimum wall	SW-WO-6

Thickness monitoring locations (SW-WO-6)

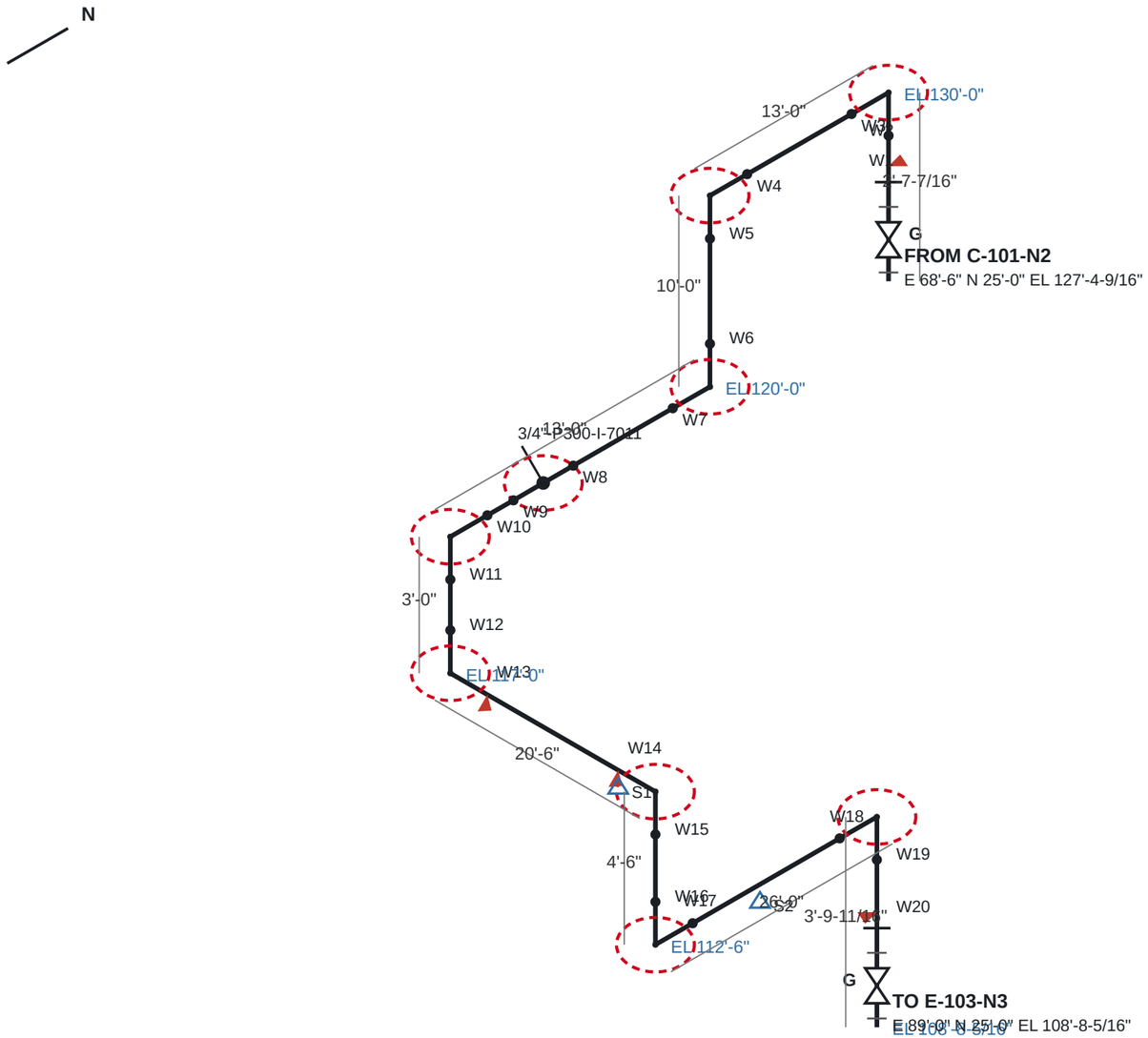
TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-200-04-TML01	6"-P300-P-1011/el8	Elbow 6"-P300-P-1011/el8 extrados	E 68'-6" N 24'-11-3/16" EL 129'-11-3/16"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML02	6"-P300-P-1011/p9	Pipe 6"-P300-P-1011/p9 mid-length	E 68'-6" N 18'-6" EL 130'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML03	6"-P300-P-1011/el10	Elbow 6"-P300-P-1011/el10 extrados	E 68'-6" N 12'-13/16" EL 129'-11-3/16"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML04	6"-P300-P-1011/p11	Pipe 6"-P300-P-1011/p11 mid-length	E 68'-6" N 12'-0" EL 125'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML05	6"-P300-P-1011/el12	Elbow 6"-P300-P-1011/el12 extrados	E 68'-6" N 11'-11-3/16" EL 120'-13/16"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML06	6"-P300-P-1011/p13	Pipe 6"-P300-P-1011/p13 mid-length	E 68'-6" N 6'-1/2" EL 120'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML07	6"-P300-P-1011/T:3/4"-P300-I-7011	Tee 6"-P300-P-1011/T:3/4"-P300-I-7011 branch outlet	E 68'-6" N 6" EL 120'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML08	6"-P300-P-1011/el16	Elbow 6"-P300-P-1011/el16 extrados	E 68'-6" N -11-3/16" EL 119'-11-3/16"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML09	6"-P300-P-1011/el18	Elbow 6"-P300-P-1011/el18 extrados	E 68'-6-13/16" N -12" EL 117'-13/16"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML10	6"-P300-P-1011/p19	Pipe 6"-P300-P-1011/p19 mid-length	E 78'-9" N -12" EL 117'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML11	6"-P300-P-1011/el20	Elbow 6"-P300-P-1011/el20 extrados	E 88'-11-3/16" N -12" EL 116'-11-3/16"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML12	6"-P300-P-1011/p21	Pipe 6"-P300-P-1011/p21 mid-length	E 89'-0" N -12" EL 114'-9"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML13	6"-P300-P-1011/el22	Elbow 6"-P300-P-1011/el22 extrados	E 89'-0" N -11-3/16" EL 112'-6-13/16"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML14	6"-P300-P-1011/p23	Pipe 6"-P300-P-1011/p23 mid-length	E 89'-0" N 12'-0" EL 112'-6"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-200-04-TML15	6"-P300-P-1011/el24	Elbow 6"-P300-P-1011/el24 extrados	E 89'-0" N 24'-11-3/16" EL 112'-5-3/16"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Gauge calibrated on a step block; operator qualification checked	Owner UT procedure	Calibration within the owner's tolerance	Calibration record	R	R	R				
2	Readings at 15 TMLs	SW-WO-6	Four readings per TML recorded	Thickness record per TML	S	S	S				
3	Results reviewed against the owner's minimum wall	Owner integrity procedure	Every reading above the minimum, or a work order raised	Signed thickness report	R	H	R				
4	Insulation and access reinstated; work order handed back	SW-ITP-3	All activities accepted	Signed ITP	H	H	R				

ISOMETRIC — 6"-P300-P-1011 — Ultrasonic thickness survey of 6"-P300-P-1011



BILL OF MATERIALS

1	GASKET 6" P300 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 6" P300	2
3	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300	4
4	FLANGE WELD-NECK 6" P300 RAISED FACE	2
5	PIPE 6" P300 BEVELLED ENDS	79'-2-13/16"
6	ELBOW 90° R = 5/3 OD 6" P300	8
7	TEE REDUCING 6" × 3/4" P300	1

WELDS 20 (4 FIELD) · SUPPORTS 2 · SPOOLS 5

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F W14F W15 W16 W17 W18 W19 W20F

6"-P300-P-1011

CLASS H300 — Hydrocarbon process
SERVICE P · 6" · P300 · DESIGN 300 psi
C-101-N2 → E-103-N3
PIPE 79'-2-13/16" · WEIGHT 1677 lb

DWG ISO-6"-P300-P-1011 REV 0 WO-200-04

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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